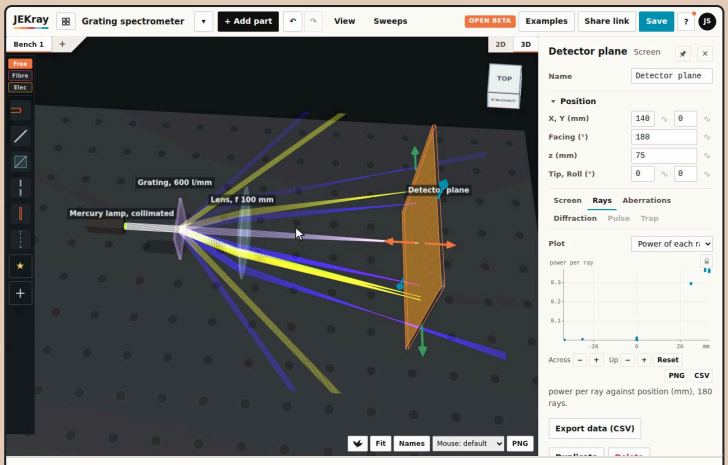
OPTICAL BENCH SIMULATOR · FREE IN YOUR BROWSER

# JEKray

## An optical bench simulator that shows the waves as well as the rays.

Lay out sources, lenses, mirrors, fibre and detectors on a virtual breadboard and see the light traced as you move them: exact rays through real glass, diffracted Gaussian beams with interference and polarisation, in 2D and 3D. Nothing to install, no account needed.



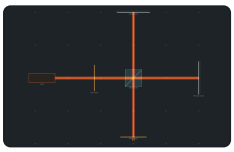

A grating spectrometer, turned in the 3D view. Drag any part and the light follows.

**Checked against published results**

**Makers' lens files**
Thorlabs and Edmund .zmx and .zar files import and focus to the makers' stated focal lengths.

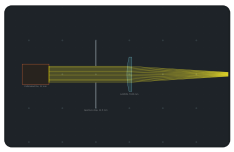
**Published results**
Fox and Li's resonator losses, Safronova's polarisabilities, measured rubidium absorption, a levitated nanosphere's damping.

**Optical design**
Real rays, Seidel aberrations and Strehl ratios of a singlet and a Cooke triplet against an independent ray tracer.



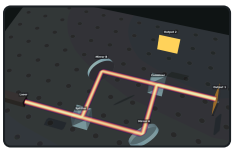
### Interference, cavities and polarisation

Each ray carries its polarisation and optical path, so interferometers show fringes and cavities resonate. The diffracted field is worked out where rays cannot tell the story: slits, masks, a tight focus, a vortex beam.



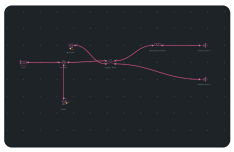
### The makers' catalogues, built in

Schott, Ohara, Hoya, CDGM and Sumita, some 1,200 glasses, plus crystals and optical plastics. Import a Thorlabs or Edmund lens file; export a prescription your lens-design program opens.



### The diffracted field in 3D

Raise, tip and roll any part and the light follows. The field glows along each beam in calibrated W/cm², interference and all. Bring mounts and housings in from CAD as STL files.



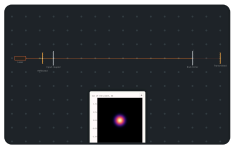
### Fibre, free space and the signals between

Fibre subsystems, free-space optics and the electronics after the detector on one bench: photodiode signals, oscilloscopes, lock-ins and lock error signals. Pulses, nonlinear optics, vapour cells and optical traps too.



### JEKrayPro: the cloud, for you or a team

Benches save as you work, with their whole history. Fork one and merge it back. A team shares one workspace with roles, invitations by email, one editor at a time and an activity record.



### 48 examples and a two-minute tour

A Michelson, a Fabry–Pérot, a grating spectrometer, optical tweezers, a PDH lock, a mode converter: open one, break it, read the results. Documentation sets out every model with its equations.

**FREE**
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The whole simulator, the examples, files and share links.

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Free, with or without an account. Prices include VAT where it applies. JEKray is in open beta and checked against published results on its validation page.

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